NAME: DHILIP SANJU V

COLLEGE: RAJALAKSHMI ENGINEERING COLLEGE

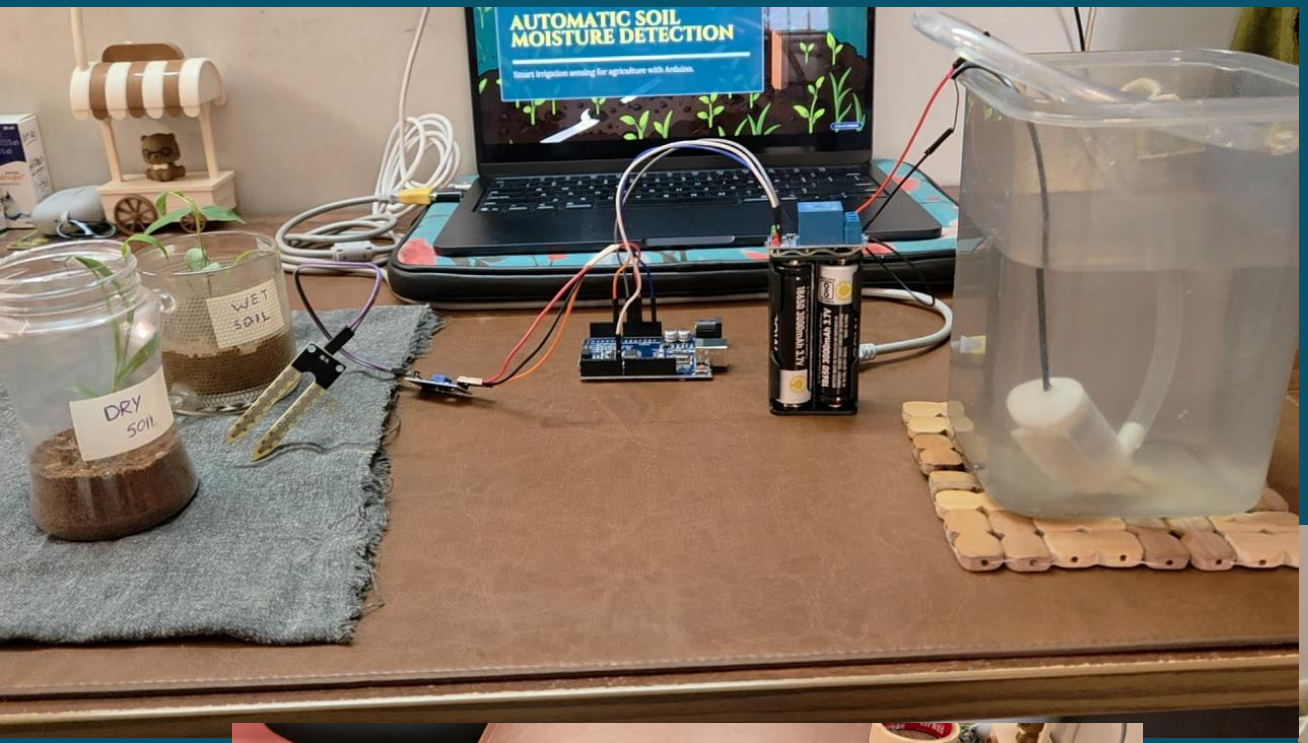
AUTOMATIC SOIL MOISTURE DETECTION

Smart irrigation sensing for agriculture with Arduino.

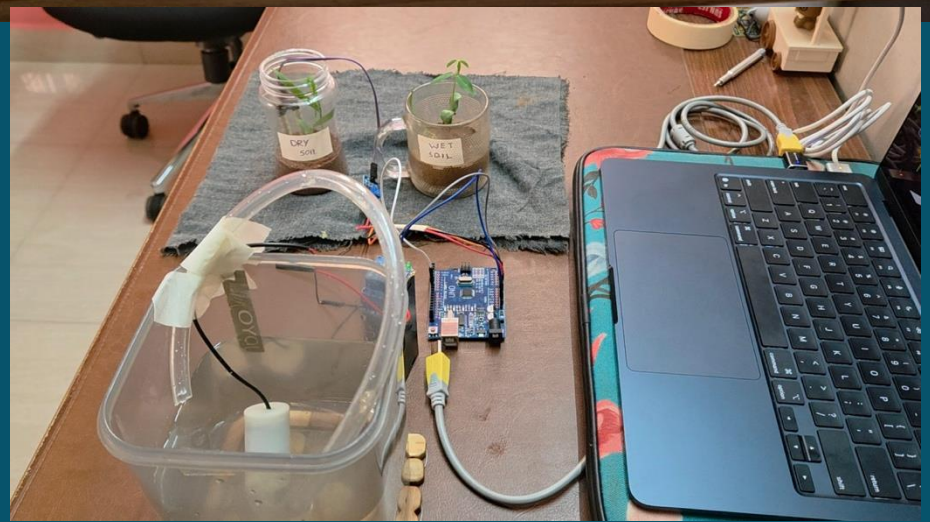


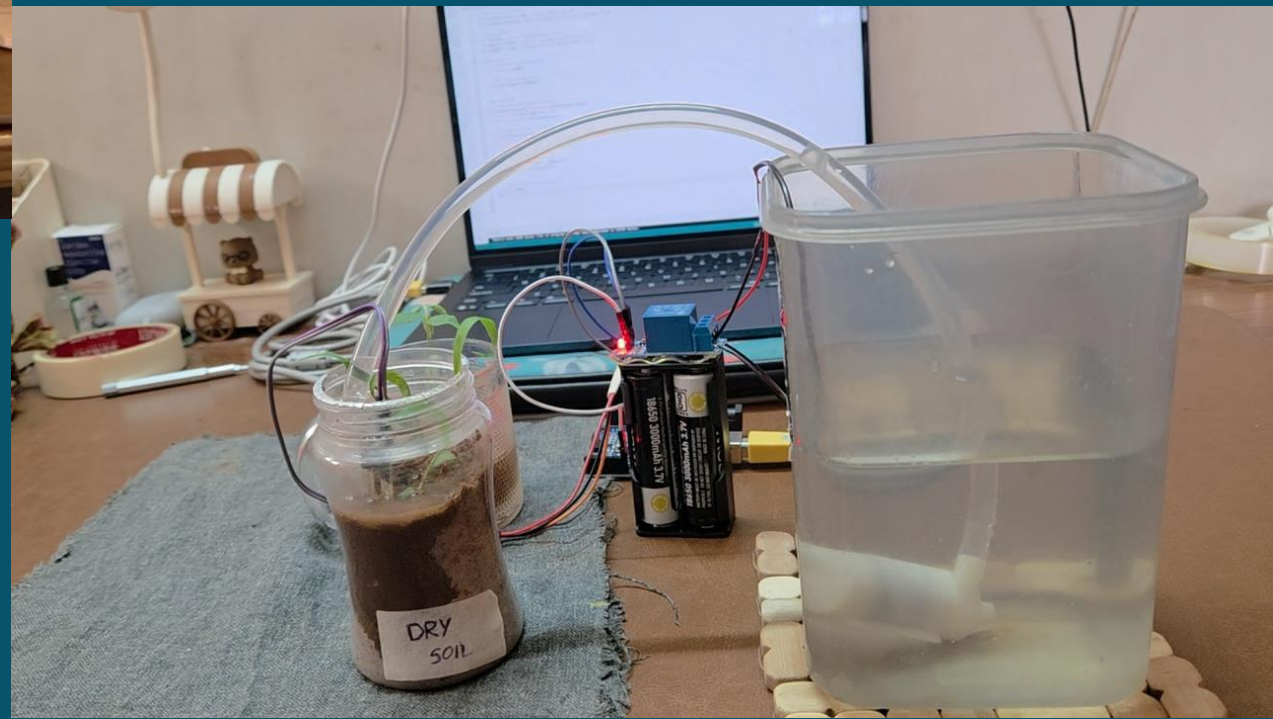
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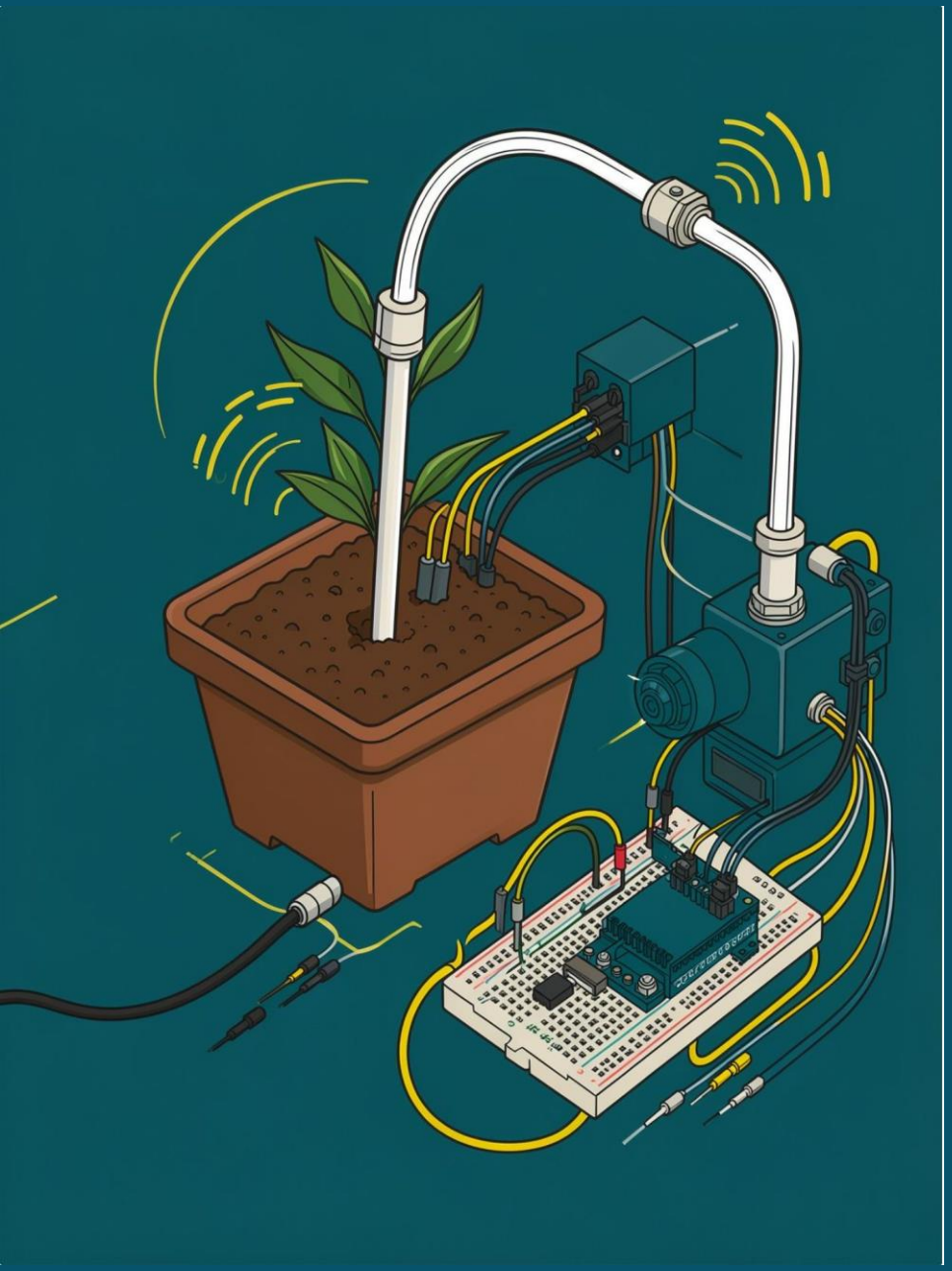
AUTOMATIC SOIL MOISTURE DETECTION









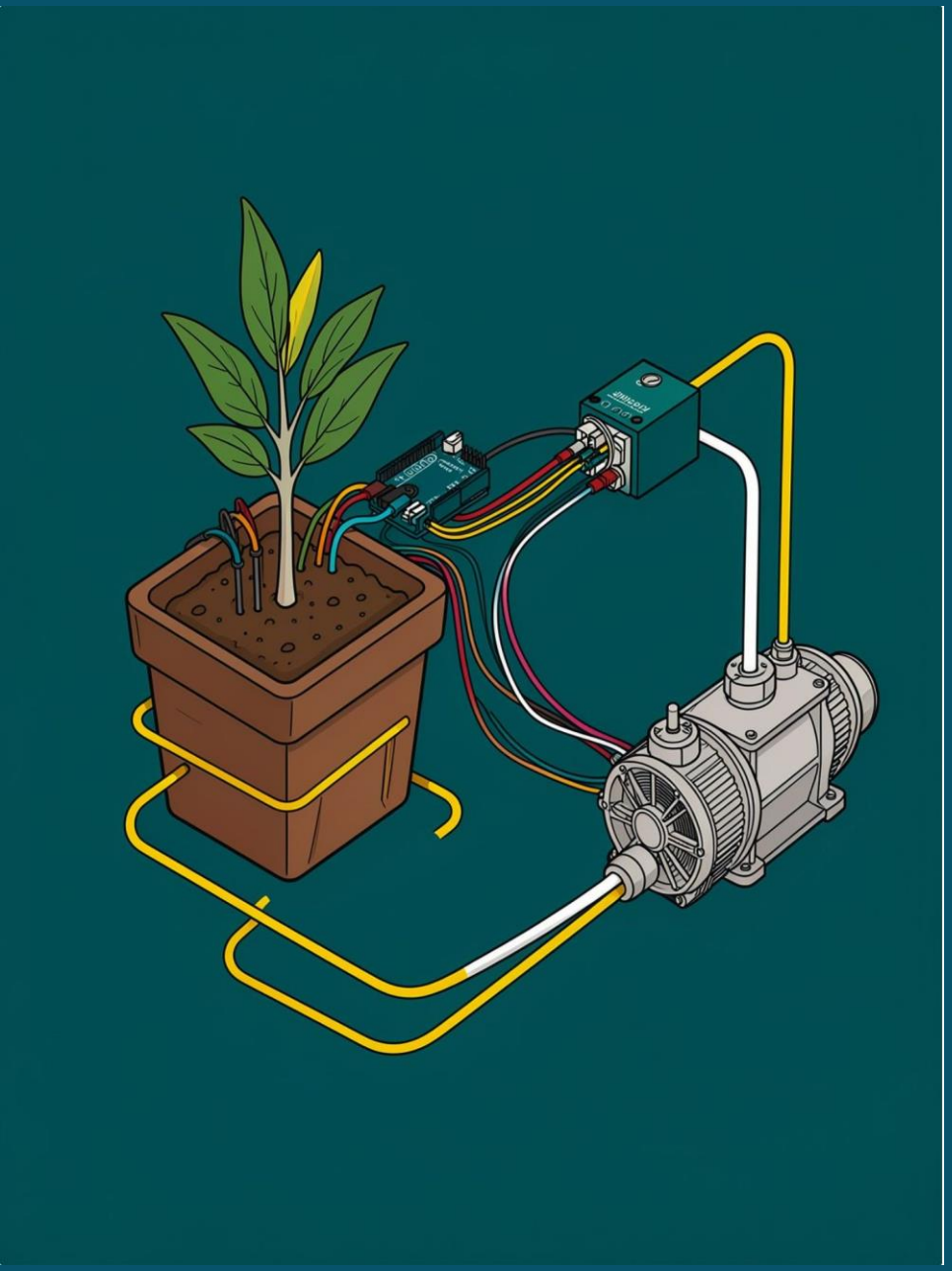


HARDWARE

PROJECT-RELATED DESCRIPTION

The Automatic Soil Moisture Detection System is designed to moniter the moisture level in the soil. It uses a soil moisture sensor to dedect whether the soil is dry or wet. An Arduino Uno processes the sensor readings and controls the water pump automatically. This system helps conserve water, reduce manual effort and pronmote efficient irrigation.

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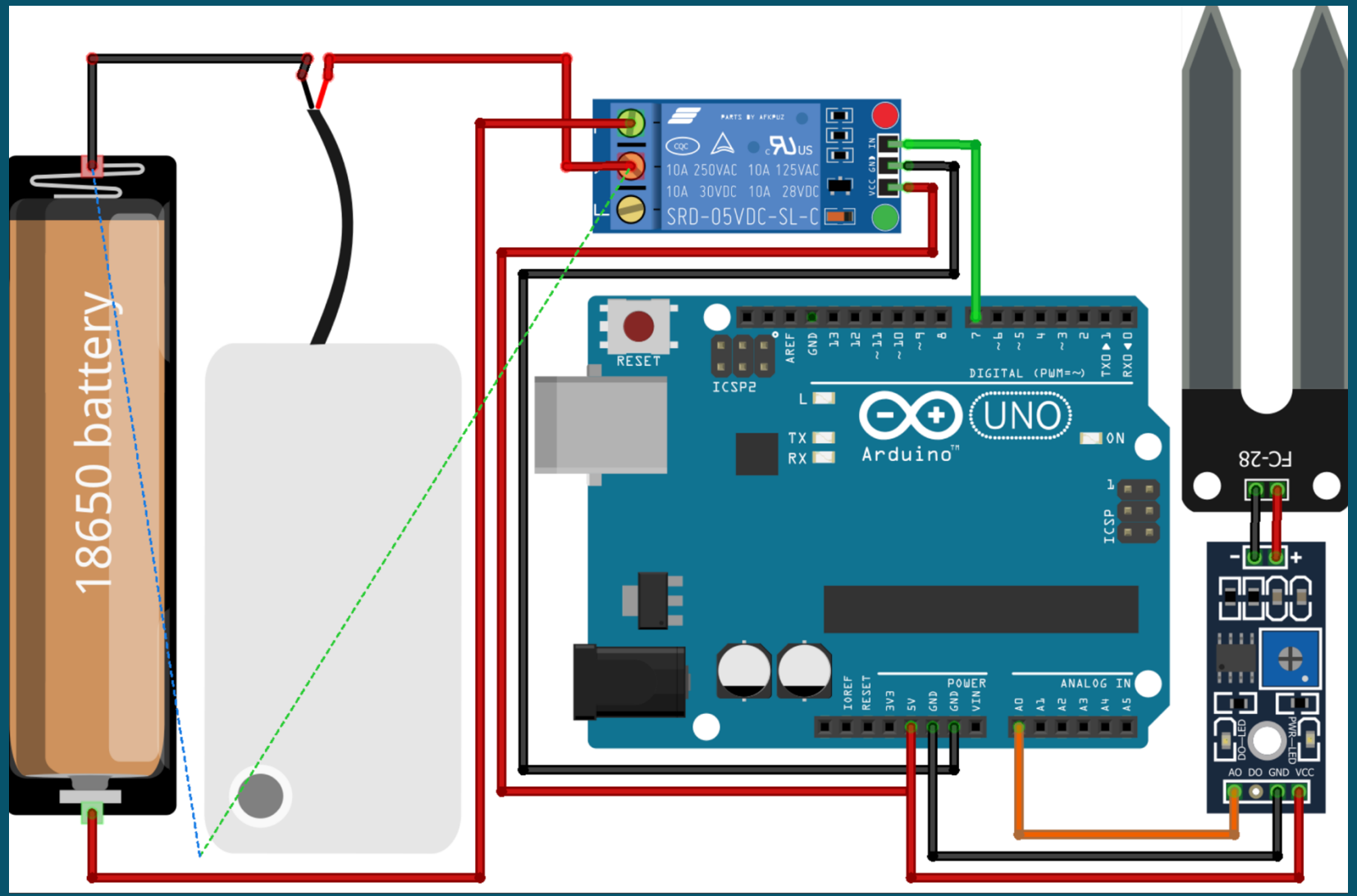
COMPONENTS REQUIRED

REQUIRED MATERIALS AND COMPONENTS

- ARDUINO UNO
- SOIL MOISTURE SENSOR
- 5V RELAY MODULE
- WATER PUMP MOTOR
- SILICONE TUBE
- BREADBOARD
- JUMPER CABLES
- 3.7V LITHIUM BATTERY
- PLANT WITH SOIL
- WATER SUPPLY FOR MOTOR

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CIRCUIT DIAGRAM



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OPERATING LOGIC

CIRCUIT AND CONNECTION EXPLANATION

- The automatic irrigation system consists of three main circuits:
    1. **Soil Moisture Sensor Circuit:** Detects the moisture level in the soil and sends signals to the Arduino Uno.
    2. **Relay and Water Pump Circuit:** Controls the water pump based on the detected moisture level.

- The Arduino processes the sensor readings and calculates the soil moisture reading.
- When the moisture level falls below the predefined threshold, the relay activates the water pump.
- When sufficient moisture is detected, the pump switches OFF automatically, helping conserve water.

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CIRCUIT CONNECTIONS

• The soil moisture sensor has three connections: VCC, GND, and AO. VCC and GND connect to Arduino 5V and GND, while AO connects to analog pin A0.

• The relay module's VCC and GND connect to the Arduino's 5V and GND terminals, while its IN pin connects to digital pin D7.

• The water pump is connected to an external power supply through the relay's COM and NO terminals.

• When the soil becomes dry, the Arduino activates the relay to start the pump. Once sufficient moisture is reached, the relay switches OFF the pump automatically.

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BUILD

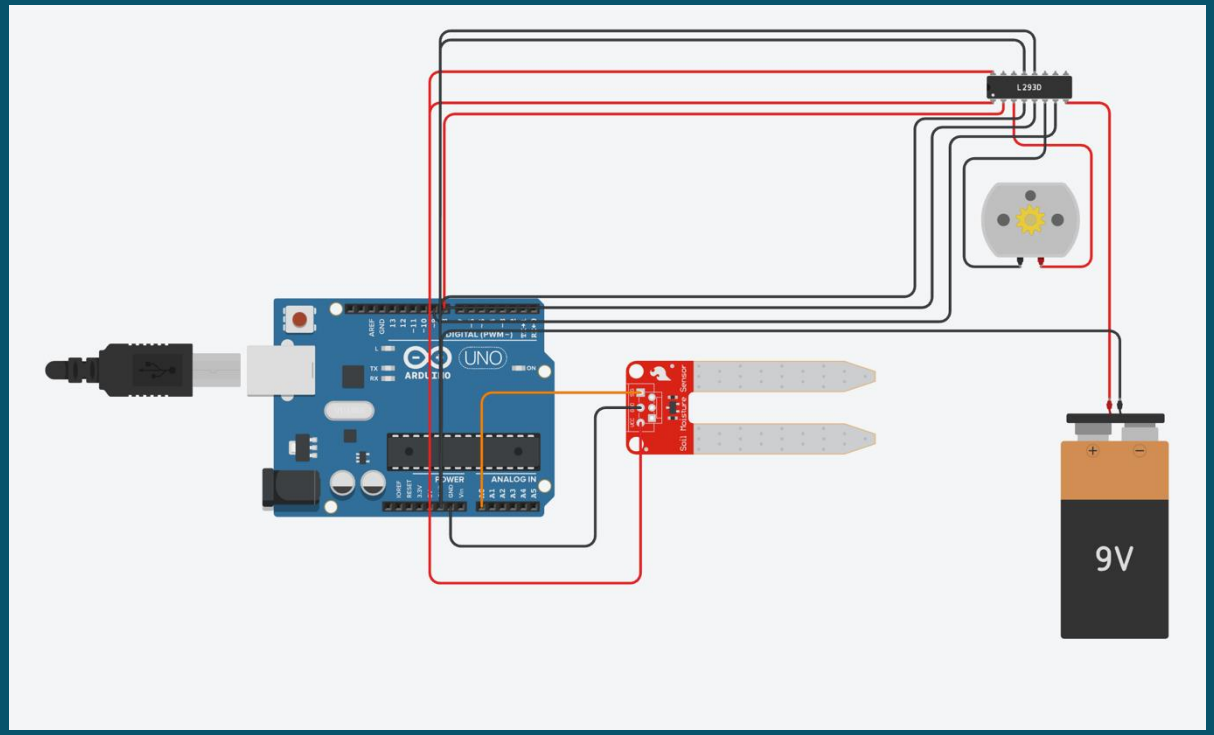
SOFTWARE- TINKERCAD

- The circuit was designed and simulated using **Tinkercad Circuits** to test the automatic irrigation system.

- The Arduino Uno reads soil moisture values. Since the water pump and relay module were unavailable, a **DC motor and L293D H-Bridge driver** were used instead.

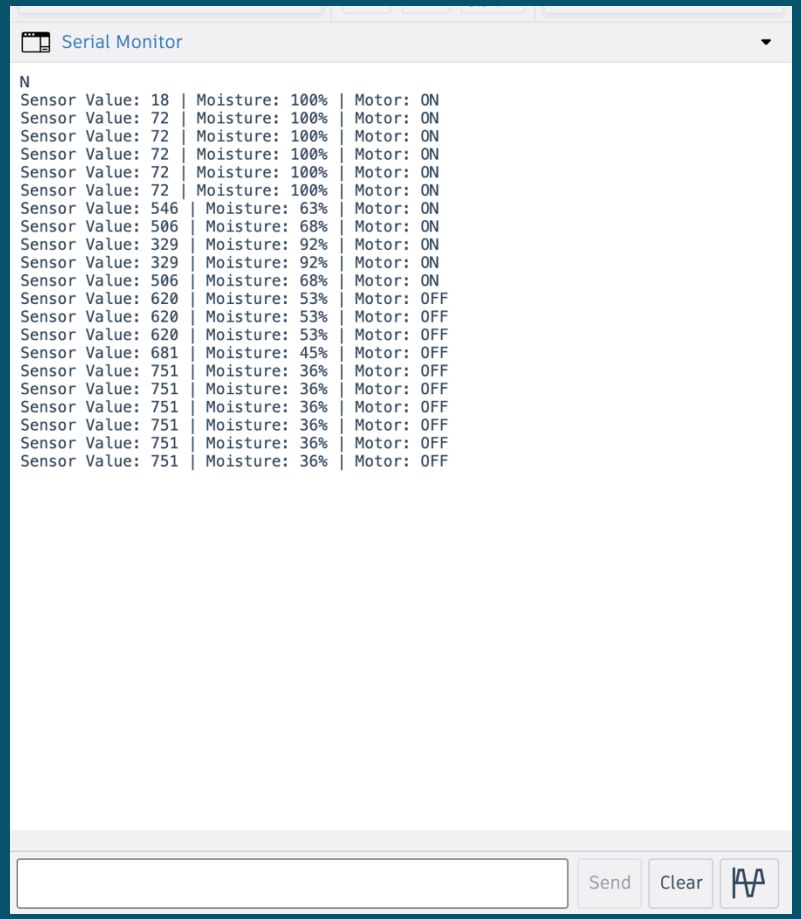
- The motor turns ON when soil moisture falls below the set threshold and turns OFF when sufficient moisture is detected.

- The simulation helped verify circuit connections, sensor readings and automatic motor control.



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TINKERCAD - SIMULATION



👈Serial Monitor readings show the sensor value and moisture percentage and the Motor condition

Tinkercad simulation testing video👇

https://drive.google.com/file/d/1wLUDQyJerZx0T_HUQ0B4qesZfvAPTf-N/view?usp=drivesdk

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ARDUINO IDE

Arduino Software

Arduino IDE was used to write and upload the program to the Arduino Uno. The system helps maintain soil moisture and reduces water wastage.

The full ready code for Automatic Soil Moisture Detection System is given below 👇

https://1drv.ms/w/c/96071CFFE6F9F70F/IQApwz0OfotcQ7fiZPEK7pvuAUgKT53ynYnjyoj1RuTvx68

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WORKING

VIDEO / SIMULATION

The working demo or simulation video is given in the link below 👇

https://drive.google.com/file/d/1mwgqxy1OlaNkDEJOHor2uYZGMLZA7k53/view?usp=drivesdk

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THANK YOU

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